

ByteBite: A Deep Learning-Based Tool for Real-Time Estimation of Food Nutritional Components to Support Diabetes Self-Management

Saim Sultan, Ismail Abu-Shanab, Husam Ghazaleh Ph.D



1. Introduction

Diabetes affects approximately 40.1 million people in the United States, 12.0% of the population and costs the U.S. healthcare system an estimated \$412.9 billion annually [2,3,4]. Tracking carbs and calories is key to managing blood glucose in diabetic patients.. Many existing applications help diabetic patients track their food intake; however, these applications are either not user-friendly (manual logging) or user-friendly, though costly. This project presents ByteBite, a deep learning based tool that estimates a meal's nutritional content (carbohydrates, protein, fat, and calories) from a single overhead smartphone photograph. The model is intended for deployment as an Android application, requiring no manual entry, targeting a platform that accounts for approximately 73% of the global mobile operating system market [5].

The App is integrated with a deep learning model that was trained on the Nutrition5k dataset, real world cafeteria dishes with high accuracy nutrition labels [1]. It comprises roughly 180 gigabytes of images, rotating videos, and comma separated metadata recording dish identifiers and nutrition information, and includes 5,006 overhead RGB dish entries across two cafeteria sites [1]. Prior work has approached this task in several ways. Nutrition5k [1] was released with baselines that regress nutrition from RGB images and, separately, from RGB combined with depth , and the publicly available SnapNutrition [6] base model applied convolutional transfer

learning to the same dataset

2. Methods

2.1 Data Cleaning and Split

Dish entries with zero recorded calories were removed, leaving 4,766 labeled dishes. Cross-referencing against available overhead RGB images yielded 3,260 dishes, and one dish with a physically implausible, mislabeled nutrition entry was removed, producing a final dataset of 3,259 dishes.

The dataset was randomly partitioned at the dish level into 70% training, 15% validation, and 15% test, giving 2,281 training, 488 validation, and 490 test dishes. The split was constructed on fixed indices and verified to contain no dish in more than one set.

2.2 Model Architectures

Three different CNN [reference] models were built and trained to identify the best-performing architecture. All models perform multi-output regression, predicting the five nutritional targets from a single RGB image. Each was trained with the Adam optimizer and a Huber loss, with targets standardized using training-set statistics.

in EfficientNetB3, we unfreeze the upper backbone blocks while keeping batch normalization layers frozen. EfficientNetV2B0 was fine-tuned in two phases, training the regression head before unfreezing the full backbone at a reduced learning rate. The Residual CNN comprises five convolutional blocks of increasing filter depth (32–512), each with ReLU activations, batch normalization, and max pooling, with residual skip connections in the deeper blocks.

Model	Backbone	Input	Head
EfficientNetB3	EfficientNetB3, ImageNet	300×300	GAP → Dense(256, ReLU) → Dropout(0.3) → Dense(5)
EfficientNetV2B0	EfficientNetV2B0, ImageNet	180×180	GAP → Dropout(0.3) → Dense(256, ReLU) → Dense(5)
Residual CNN	Trained from scratch	180×180	Flatten → Dropout(0.4) → Dense(256) → Dropout(0.3) → Dense(128) → Dense(5)

Table 1. Model Architecture

2.3 Evaluation

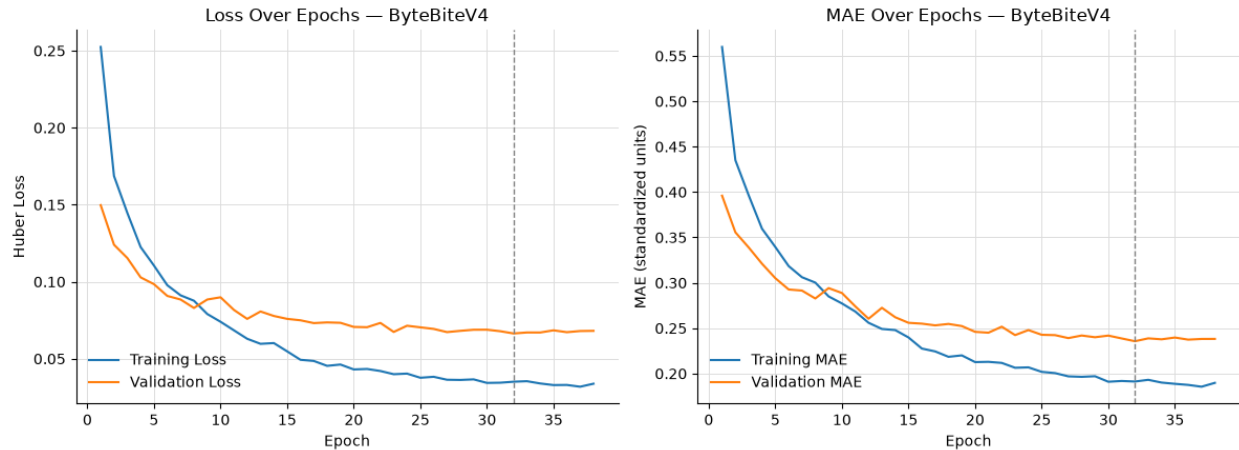
Model performance was measured using Mean Absolute Error (MAE) for each of the five nutritional targets. To account for variation introduced by random weight initialization, each model was trained across five independent runs, and 95% confidence intervals were computed for each target using bootstrap

3. Results

EfficientNetB3 achieved the lowest MAE across all five nutritional targets, with the largest gains seen for carbohydrate and protein . EfficientNetV2B0 ranked second, followed by the Residual CNN — all three outperformed the SnapNutrition baseline The two best-performing models employ ImageNet pretraining, indicating that transfer learning is well suited to a training set of this size. EfficientNetB3 s higher input resolution and orientation augmentation appear to resolve the finer visual cues that distinguish carbohydrate and protein content. As carbohydrate intake is one of the determinants of postprandial glycemic response and insulin dosing, accuracy on this target is of particular relevance to the main objective of the App

Model	Calories MAE (kcal)	Mass MAE (g)	Carb MAE (g)	Protein MAE (g)	Fat MAE (g)
EfficientNetB3	42.02 (95% CI: 40.20–44.00)	27.00 (95% CI: 26.00–28.06)	4.12 (95% CI: 3.95–4.28)	4.29 (95% CI: 4.03–4.54)	3.17 (95% CI: 3.02–3.31)
EfficientNetV2B0	48.99 (95% CI: 46.58–51.31)	31.84 (95% CI: 30.54–33.29)	7.42 (95% CI: 7.14–7.69)	6.72 (95% CI: 6.38–7.04)	4.41 (95% CI: 4.20–4.61)
Residual CNN	49.63 (95% CI: 47.38–51.86)	34.02 (95% CI: 32.49–35.68)	7.92 (95% CI: 7.64–8.22)	7.09 (95% CI: 6.71–7.52)	4.70 (95% CI: 4.50–4.91)
SnapNutrition (baseline)	101.89 (95% CI: 98.48–105.36)	69.23 (95% CI: 66.64–71.70)	10.47 (95% CI: 10.17–10.79)	11.29 (95% CI: 10.84–11.72)	8.33 (95% CI: 8.00–8.65)

Table 2. Mean absolute error per nutrient (mean and 95% bootstrap confidence interval over 1000 resamples of pooled test predictions from five independent runs).



(Figure 1. Training and validation loss and mean absolute error per epoch for the fine-tuned EfficientNetB3 model. Mean absolute error is shown in standardized target units)

4. Conclusion

In this project, we proposed ByteBite, a deep learning-based tool that estimates a meal's nutritional content (carbohydrates, protein, fat, and calories) from a single overhead smartphone photograph. The app leverages AI to help diabetic patients track their food with minimal burden and effort. Three CNN models were built and trained on the Nutrition5k dataset. EfficientNetB3 achieved the lowest MAE across all five nutritional targets, with the largest gains seen for carbohydrate and protein. These results suggest that EfficientNetB3 offers a promising foundation for accurate, low-effort dietary tracking in diabetes management

5. Future Work

The Nutrition5k dataset includes overhead depth images that were not utilized in this work; incorporating depth as an additional input could improve portion size estimation, a known source of calorie and mass error. Furthermore, the app could be extended with additional predictive models, such as one estimating blood glucose response following patient activity (food intake or exercise) and another estimating HbA1c, so that dietary estimates contribute directly to glycemic management rather than serving as isolated measurements.

References

- [1] Q. Thames, A. Karpur, W. Norris, F. Xia, L. Panait, T. Weyand, and J. Sim, "Nutrition5k: Towards Automatic Nutritional Understanding of Generic Food," arXiv:2103.03375, 2021.
- [2] Centers for Disease Control and Prevention, "National Diabetes Statistics Report," <https://www.cdc.gov/diabetes/php/data-research/index.html>, 2023.
- [3] National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), "Diabetes Statistics," <https://www.niddk.nih.gov/health-information/health-statistics/diabetes-statistics>,

2024

[4] American Diabetes Association, "Economic Costs of Diabetes in the U.S. in 2022," Diabetes Care, vol. 47, no. 1, pp. 26-43,
<https://diabetesjournals.org/care/article/47/1/26/153797/Economic-Costs-of-Diabetes-in-the-U-S-in-2022>,
2024

[5] "Android Global Market Share Statistics 2026," CommandLinux
<https://commandlinux.com/android/android-global-market-share-statistics/>, 2026

[6] D. Yerosu, "AC215_snapnutrition: Nutrition5k EDA Base Model," GitHub repository,
https://github.com/dyerosu/AC215_snapnutrition/blob/main/notebooks/Nutrition5k_EDA_Base_Model.ipynb,
accessed 2026.